



Context-Aware Vehicle Horn Detection and Violation Classification Using IoT Telemetry and YOLO-Based Visual Justification



Author: Aryan Agrawal (12312973)

1. INTRODUCTION

Noise pollution from unnecessary vehicle honking is a critical public health issue, particularly in silence zones around hospitals, schools, and courts.

Conventional microphone-based approaches cannot reliably attribute a honk to a specific vehicle and may raise privacy concerns through ambient audio capture. They are also susceptible to false positives from sirens, music, and other environmental sounds.

We propose an IoT + Computer Vision system addressing three key gaps: (1) vehicle-specific horn detection through non-destructive in-circuit current sensing, (2) real-time GPS geofence monitoring, and (3) context-aware honk justification using YOLO-based pedestrian detection.



2. OBJECTIVES

- Develop a non-destructive, vehicle-specific horn sensing module using an ACS712 Hall-effect current sensor integrated through a fuse-box adapter, avoiding wire cutting or splicing.
- Develop an ESP32-based IoT telemetry system to fuse horn state and GPS data and transmit structured telemetry over MQTT.
- Implement reliable silence-zone monitoring using GPS quality filtering, polygon-based geofencing, and debounced state transitions.
- Develop context-aware honk classification using YOLO-based pedestrian detection to distinguish potential safety honks from violations within a defined contextual window.
- Generate structured event records containing location, horn state, geofence status, and classification results for subsequent analysis and enforcement.

3. LIMITATIONS OF EXISTING APPROACHES



AUDIO / MICROPHONE BASED

- Captures ambient noise
- High risk of False positives
- Raises privacy concerns



SMARTPHONE AUDIO BASED

- Requires access to microphone
- Requires a smartphone to remain available during operation
- Ambient speech and other sounds become part of sensing environment again raises privacy concerns
- Primarily focuses on honk detection, rather than contextual justification and vehicle-specific attribution

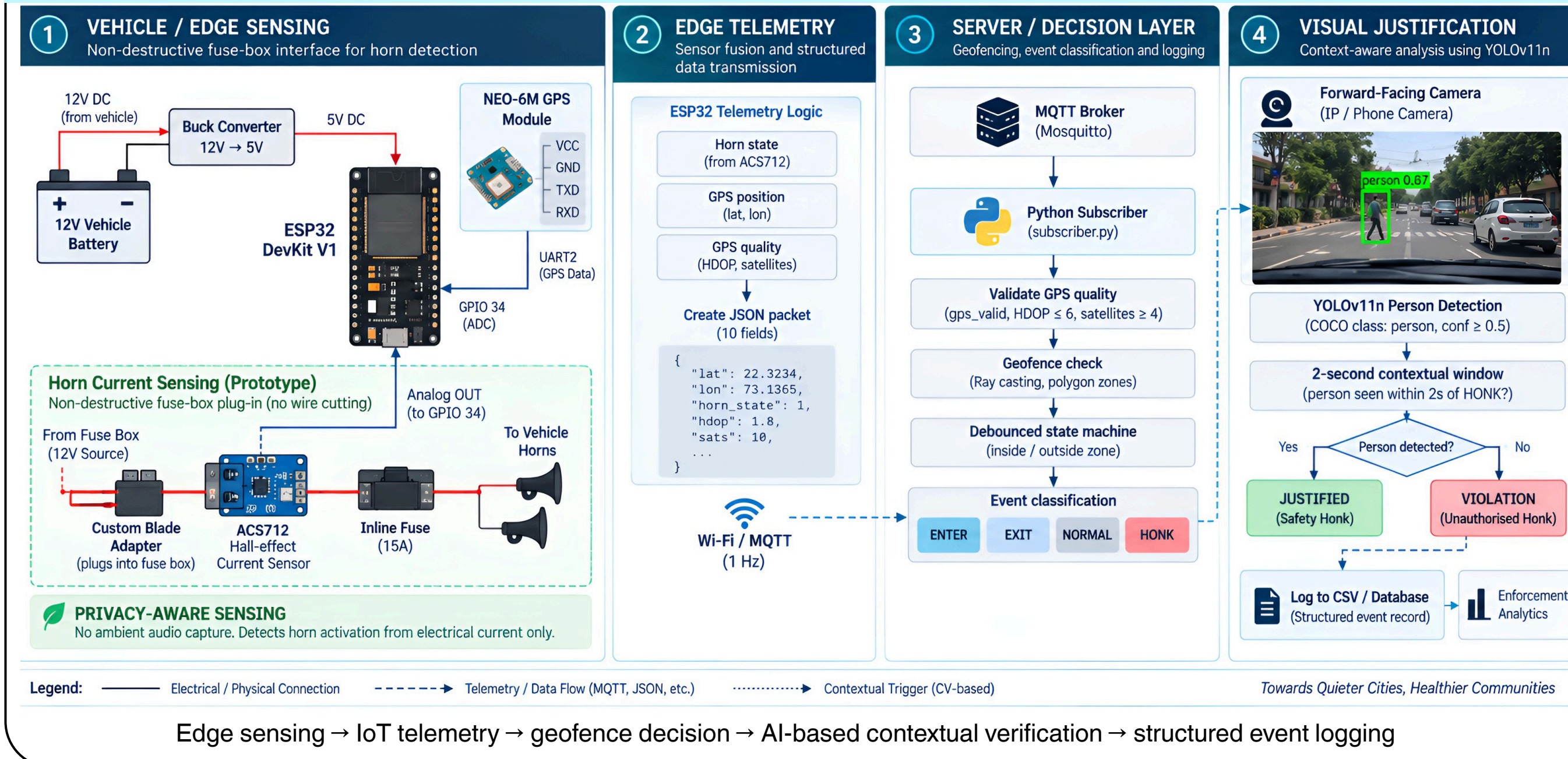


DIRECT ELECTRICAL BASED

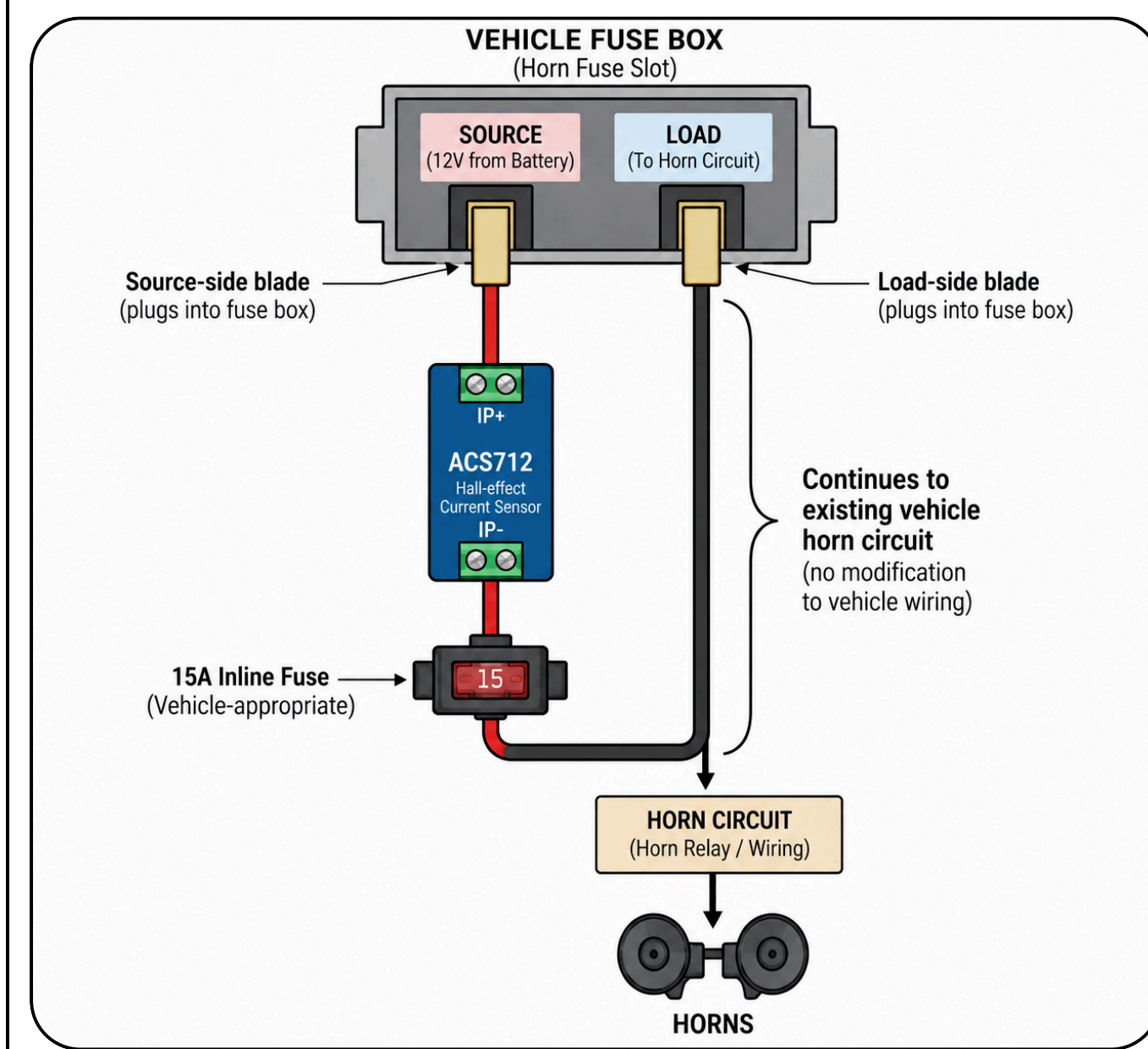
- Conventional implementations may require direct modification of the vehicle's horn circuitry
- Can increase installation complexity and reduce portability

4. METHODOLOGY

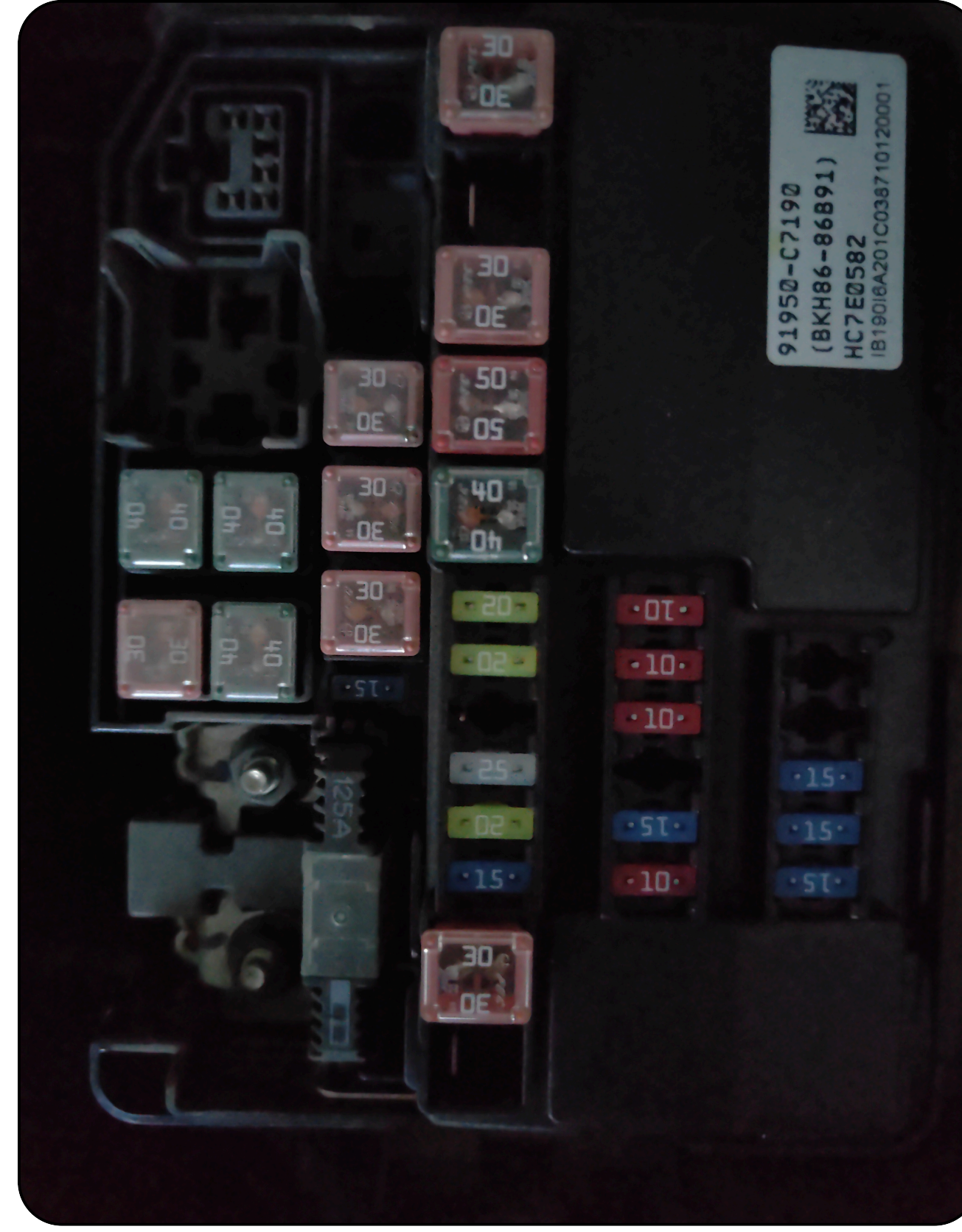
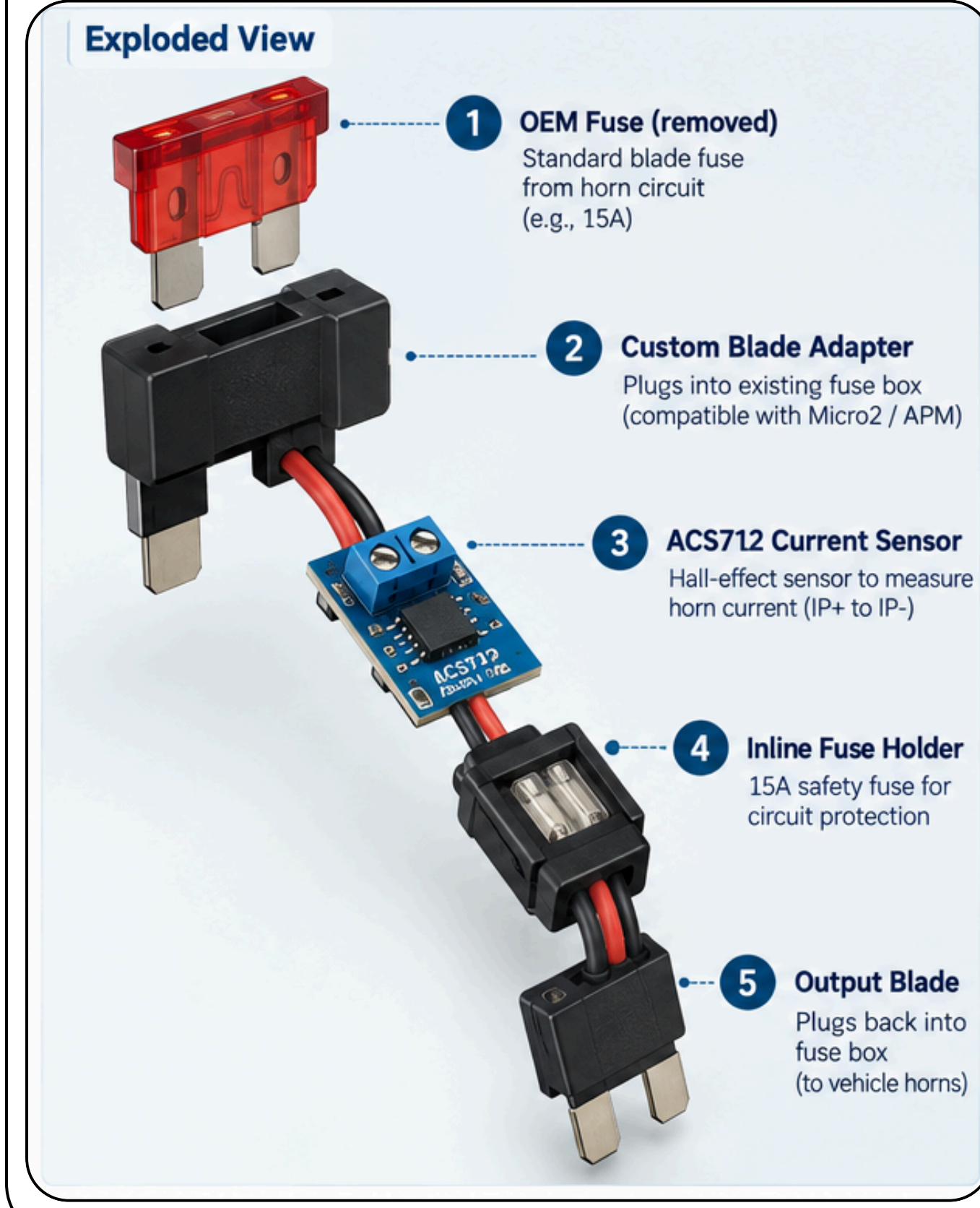
4.1 SYSTEM WORKFLOW & ARCHITECTURE



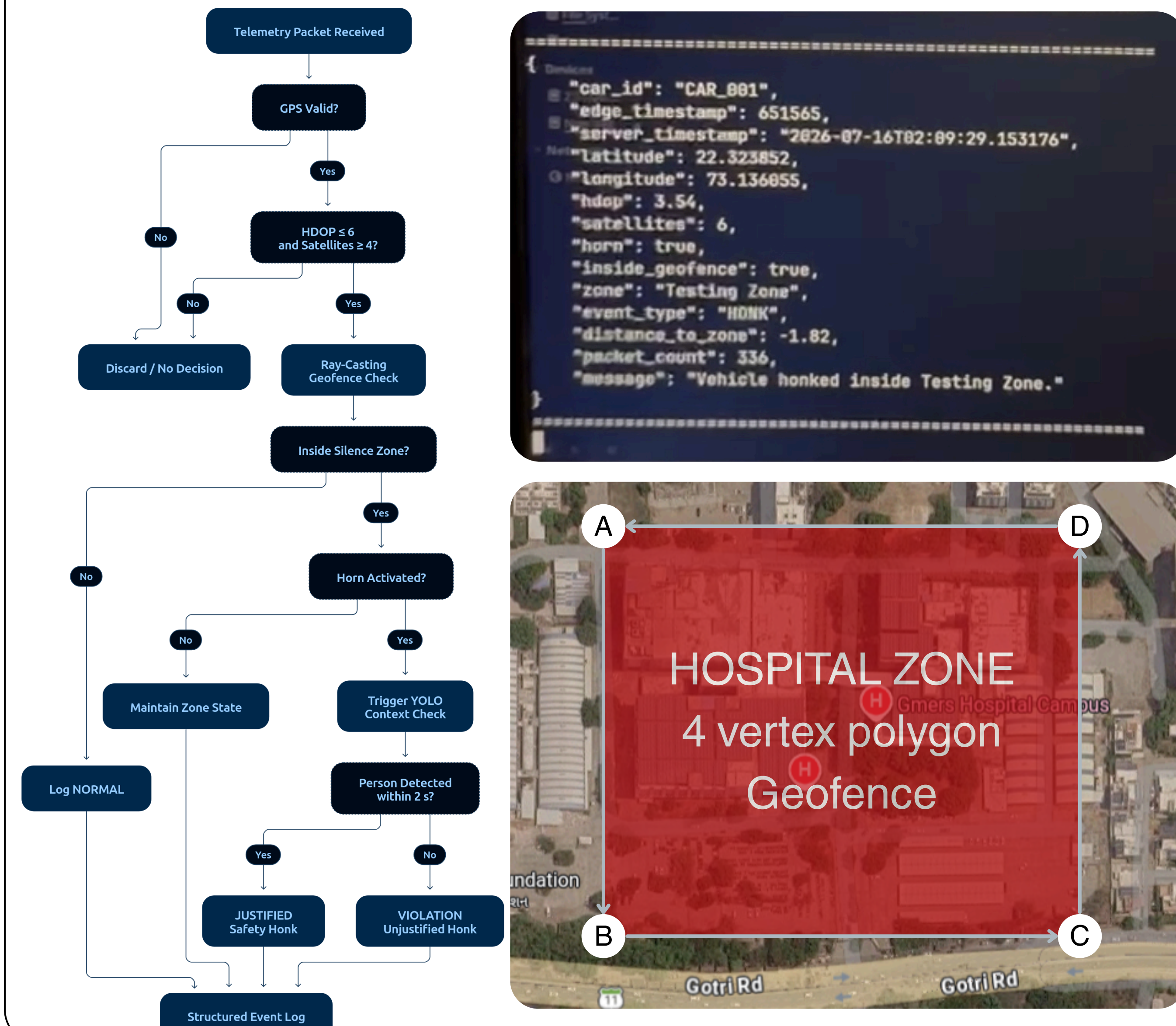
4.2 NOVEL NON-DESTRUCTIVE HORN-SENSING ARCHITECTURE



- Custom fuse-box interface enables in-circuit horn-current sensing without cutting or splicing the vehicle's original wiring.
- ACS712 Hall-effect current sensor detects horn activation directly from current flow in the horn circuit.
- Inline module ensures maximum accuracy.
- Plug-in architecture allows portability across different vehicles.
- Privacy-aware sensing: detects horn activation electrically without capturing ambient audio.



4.3 EVENT DECISION LOGIC



5. RESULTS & FINDINGS

The system was field-tested in Vadodara, Gujarat, with a vehicle driving through a defined test silence zone (polygon: 4 vertices, near coordinates 22.3237°N, 73.1361°E)

Metric	Value
Total telemetry packets collected	~3,500+
Test sessions conducted	8+
MQTT publish rate	1 packet/second
GPS quality filtering	HDOP ≤ 6.0, satellites ≥ 4
Geofence events detected	ENTER, EXIT, HONK, NORMAL
HONK events logged	358 across test sessions
EXIT transitions logged	73 across test sessions
Debounce accuracy	Zero false state transitions observed
YOLO model	YOLOv11n, person class (COCO #0)
YOLO confidence threshold	≥ 0.5
Justification window	2.0 seconds

OUTPUT FORMAT JSON PACKET

<pre>{ "car_id": "CAR_001", "latitude": 22.323741, "longitude": 73.135891, "horn": true, "inside_geofence": true, "zone": "Testing Zone", "event_type": "HONK", "justified": false, "distance_to_zone": -3.72, "message": "Honk unjustified"}</pre>	<pre>{ "timestamp": 318958, "car_id": "CAR_001", "lat": 22.323350, "lon": 73.136290, "horn": true, "hdop": 1.6, "satellites": 4, "adc": 0, "gps_age": 612, "gps_valid": true}</pre>
---	---

6. LIMITATIONS

- Full vehicle fuse-box integration is in process and currently in prototype stage
- Not enough tests conducted for edge cases
- CV inference runs on a laptop (server-side); edge deployment planned
- Single-class detection (person); extending to vehicles, animals, and obstacles

7. CONCLUSION

The proposed system combines non-destructive in-circuit horn sensing, GPS-based geofencing, IoT telemetry, and YOLO-based contextual verification to enable vehicle-specific and location-aware horn-event classification. The prototype was validated using real vehicle telemetry collected in Vadodara, demonstrating an end-to-end pipeline from horn activation to structured event classification.

Vehicle-specific sensing + geofence awareness + visual context → context-aware horn enforcement

8. REFERENCES

- [1] R. Banerjee and A. Sinha, "Two Stage Feature Extraction using Modified MFCC for Honk Detection," 2012 International Conference on Communications, Devices and Intelligent Systems (CODIS), pp. 97–100, IEEE, 2012.
- [2] B. Maity, A. Chokhani, S. Nandi, and S. Bhattacharjee, "Real-Time Car Honk Detection for On-the-Go Application using Lightweight Deep Learning Model," 2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), IEEE, 2024. DOI: 10.1109/ANTS63515.2024.10898423.
- [3] J. Dutta, D. Chakraborty, S. Roy, and D. Puthal, "JUHD: IoT-based Honk Detection and Novel Driver Profiling for Smart City," 2023 IEEE International Conference on High Performance Computing & Communications, Data Science & Systems, Smart City & Dependability in Sensor, Cloud & Big Data Systems & Application, 2023. DOI: 10.1109/HPCC-DSS-SmartCity-DependSys60770.2023.00139.
- [4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
- [5] Allegro MicroSystems, "ACS712: Fully Integrated, Hall-Effect-Based Linear Current Sensor," Datasheet.

SCAN FOR PROJECT

